

Section A — MCQs

No.	Correct Option	Answer	Reason
i	B	Formula units	NaCl is ionic, so we say formula units , not molecules. $0.1 \text{ mol} = 6.02 \times 10^{22}$ formula units.
ii	C	Broader curve shifted right	At 500 K, molecules have higher average kinetic energy, so curve becomes lower, broader, and shifted to right .
iii	C	$\frac{1}{2}, 1, \frac{3}{2}$	Rate = $k[A]^{1/2}[B]$. Order w.r.t. A = 1/2, B = 1, total = $\frac{1}{2} + 1 = \frac{3}{2}$.
iv	D	Greater nuclear force of attraction of oxygen	Oxygen has greater effective nuclear charge than nitrogen, so it attracts an extra electron more strongly.
v	B	Endothermic	Products have greater energy than reactants, so energy is absorbed.
vi	D	22414 cm^3	1 mole gas at STP occupies $22.414 \text{ dm}^3 = 22414 \text{ cm}^3$.
vii	B	Electrophilic addition	Propene has a double bond; bromine adds across it by electrophilic addition.
viii	B	Fehling's solution	Aldehydes reduce Fehling's solution to red Cu_2O .
ix	B	$\text{CH}_3 + \text{Cl} \cdot \rightarrow \text{CH}_3\text{Cl}$	Termination occurs when free radicals combine to form a stable molecule.
x	C	Square planar	XeF_4 has 4 bond pairs and 2 lone pairs; shape is square planar.
xi	D	ns^2np^5	Halogens have this configuration and are the most electronegative elements.
xii	D	AlCl_3	Al^{3+} hydrolyses water and produces acidic solution.
xiii	C	CH_3NHCH_3	Secondary amines are generally more basic than primary and tertiary amines in aqueous solution.
xiv	C	NH_3	NH_3 has a lone pair and can donate it, so it is a nucleophile.
xv	D	LiAlH_4	Ethanal is reduced to ethanol by LiAlH_4 .
xvi	B	Stronger London dispersion forces in I_2	Iodine has larger electrons/cloud size, so stronger dispersion forces make it solid.
xvii	D	Propanone	Iodoform test is given by methyl ketones, $\text{CH}_3\text{CO}-$. Propanone is CH_3COCH_3 .

Final MCQ Key

i-B, ii-C, iii-C, iv-D, v-B, vi-D, vii-B, viii-B, ix-B, x-C, xi-D, xii-D, xiii-C, xiv-C, xv-D, xvi-B, xvii-D

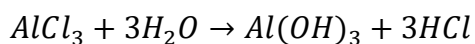
Section B — Short Questions

Q2(i)

Main: Reaction of chlorides with water + pH

a) $AlCl_3$

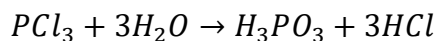
$AlCl_3$ hydrolyses in water:



Due to formation of HCl , solution is **acidic**.

pH: less than 7.

b) PCl_3



Both acids are formed, so solution is **acidic**.

pH: less than 7.

OR: Shapes of orbitals

3s orbital

Spherical shape.

$3p_x$ orbital

Dumb-bell shape along x-axis.

$3d_{z^2}$ orbital

Dumb-bell with a ring/doughnut around the middle.

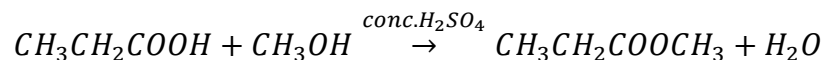
Q2(ii)

Main: Preparation of methyl propanoate

Methyl propanoate is an ester.

Alcohol: **methanol**

Carboxylic acid: **propanoic acid**



This is **esterification**.

OR: Amorphous vs crystalline solids

Crystalline solids

Amorphous solids

Regular arrangement of particles Irregular arrangement

Sharp melting point

No sharp melting point

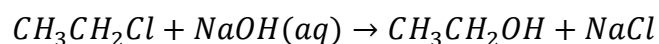
Anisotropic

Isotropic

Q2(iii)

Main: Chloroethane with NaOH

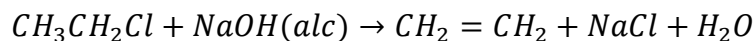
1) With aqueous NaOH



Product: **ethanol**

Reaction type: **nucleophilic substitution**

2) With alcoholic NaOH

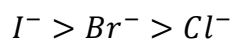


Product: **ethene**

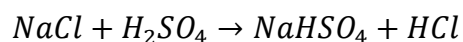
Reaction type: **elimination**

OR: Reducing power of halide ions

Reducing power increases down the group:

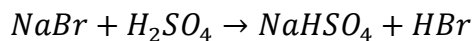


Cl^- with conc. H_2SO_4

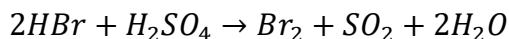


Only acid-base reaction occurs. Cl^- is not strong enough to reduce H_2SO_4 .

Br^- with conc. H_2SO_4



Then:

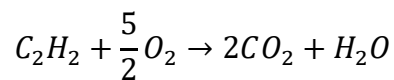


Br^- reduces H_2SO_4 , so bromide is a stronger reducing agent than chloride.

Q2(iv)

Main: Enthalpy change

Reaction:



Formula:

$$\Delta H = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

$$\Delta H = [2(-393.5) + (-285.5)] - [(227.3)]$$

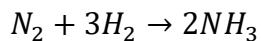
$$\Delta H = [-787 - 285.5] - 227.3$$

$$\Delta H = -1072.5 - 227.3$$

$$\Delta H = -1299.8 \text{ kJmol}^{-1}$$

So reaction is **exothermic**.

OR: Mass of NH_3 from 20 g N_2



Molar mass of $N_2 = 28g/mol$

$$moles = \frac{20}{28} = 0.714$$

1 mol N_2 gives 2 mol NH_3

$$0.714 \times 2 = 1.428 \text{ mol } NH_3$$

Molar mass of $NH_3 = 17g/mol$

$$mass = 1.428 \times 17 = 24.28g$$

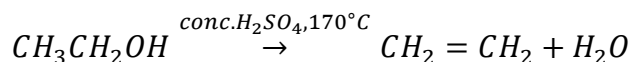
Answer: 24.28g of NH_3

Q2(v)

Main: Preparation of ethene

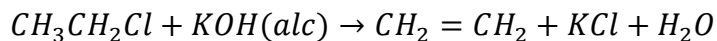
a) From ethanol

Ethanol is dehydrated using conc. H_2SO_4 at about $170^\circ C$.



b) From chloroethane

Chloroethane reacts with alcoholic KOH/NaOH.



This is elimination.

OR: Intermolecular forces

Molecule Force

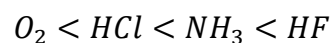
O_2 London dispersion forces

HCl Dipole-dipole forces

NH_3 Hydrogen bonding

HF Strong hydrogen bonding

Ascending order of strength:

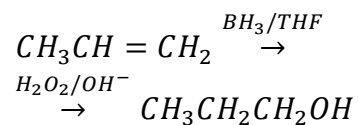


Q2(vi)

Main: Preparation of propan-1-ol

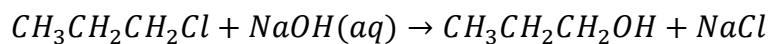
a) From propene

Hydroboration-oxidation gives propan-1-ol.



Reason: anti-Markovnikov addition gives primary alcohol.

b) From 1-chloropropane



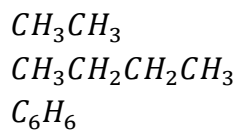
This is nucleophilic substitution.

OR: Catenation

Catenation is the ability of an element to form covalent bonds with itself.

Carbon shows strong catenation because C-C bond is strong. Therefore carbon forms chains, branched chains and rings.

Examples:



This increases the diversity of organic compounds.

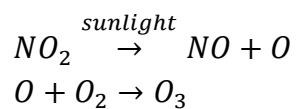
Q2(vii)

Main: Role of NO_x in PAN formation

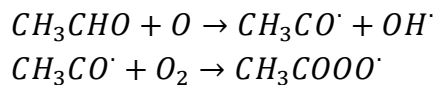
PAN means **peroxyacetyl nitrate**.

NO_x helps in photochemical smog and formation of PAN.

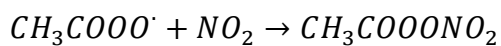
Basic steps:



Hydrocarbons form acetyl/peroxyacetyl radicals:



Then:

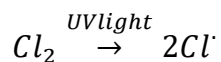


Product is **PAN**.

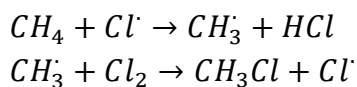
OR: Formation of CH_3Cl from methane and chlorine

This is free radical substitution.

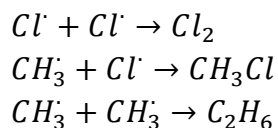
1) Initiation



2) Propagation



3) Termination

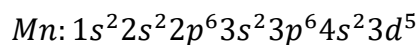


Q2(viii)

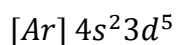
Main: Electronic configuration

a) ${}^{55}_{25}Mn$

Atomic number = 25



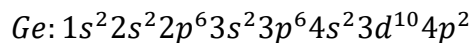
or



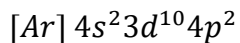
b) ${}^{64}_{32}P$

The symbol seems misprinted because atomic number 32 is **Ge**, not P.

For atomic number 32:



or



OR: Two ways to recognize equilibrium

1. The rate of forward reaction becomes equal to the rate of reverse reaction.

$$rate_f = rate_r$$

2. Concentrations of reactants and products become constant with time.
-

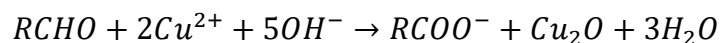
Q2(ix)

The structures A and B are unclear in the extracted text, but this question usually asks to distinguish **aldehyde and ketone**.

Chemical test: Fehling's test

Aldehyde

Aldehyde gives brick-red precipitate of Cu_2O .



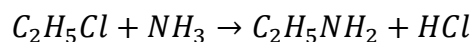
Ketone

Ketone gives no reaction.

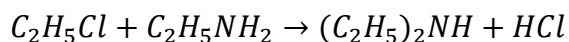
OR: Preparation of diethylamine

a) By nucleophilic substitution

Ethyl chloride reacts with ammonia/ethylamine.

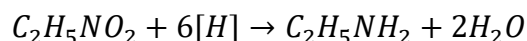


Then:

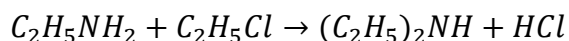


b) By reduction

Diethyl nitrosamine/amides may give secondary amines, but the standard preparation is reduction of nitroethane to ethylamine:



Then ethylamine can be alkylated:



Q2(x)

Main: Relative isotopic mass of fourth isotope

Given:

Average atomic mass = 140.116

Known isotopes:

$$^{136}\text{Ce} = 135.91, \text{abundance} = 0.185\%$$

$$^{138}\text{Ce} = 137.91, \text{abundance} = 0.25\%$$

$$^{140}\text{Ce} = 139.91, \text{abundance} = 88.45\%$$

Fourth abundance:

$$100 - (0.185 + 0.25 + 88.45) = 11.115\%$$

Let mass of fourth isotope = x

$$140.116 = \frac{135.91(0.185) + 137.91(0.25) + 139.91(88.45) + x(11.115)}{100}$$

$$14011.6 = 25.14335 + 34.4775 + 12375.0395 + 11.115x$$

$$14011.6 = 12434.66035 + 11.115x$$

$$1576.93965 = 11.115x$$

$$x = 141.875$$

Answer: relative isotopic mass $\approx$ 141.88

OR: K_{sp} of PbCl_2

Solubility = 0.5 g dm^{-3}

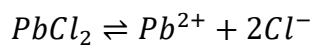
Molar mass of PbCl_2 :

$$207 + 2(35.5) = 278 \text{ g/mol}$$

Molar solubility:

$$s = \frac{0.5}{278} = 1.798 \times 10^{-3} \text{ mol dm}^{-3}$$

Dissociation:



$$K_{sp} = [\text{Pb}^{2+}][\text{Cl}^-]^2$$

$$K_{sp} = s(2s)^2 = 4s^3$$

$$K_{sp} = 4(1.798 \times 10^{-3})^3$$

$$K_{sp} = 2.33 \times 10^{-8}$$

Q2(xi)

Main: Quantum numbers for 6f orbital

For 6f:

Principal quantum number

$$n = 6$$

Azimuthal quantum number

For f orbital:

$$l = 3$$

Magnetic quantum number

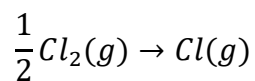
$$m_l = -3, -2, -1, 0, +1, +2, +3$$

OR: Definitions

a) Standard enthalpy of atomization

It is the enthalpy change when one mole of gaseous atoms is formed from an element in its standard state.

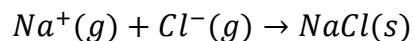
Example:



b) Standard lattice enthalpy

It is the enthalpy change when one mole of ionic crystal is formed from gaseous ions.

Example:



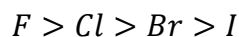
Q2(xii)

Main: Electronegativity and trend

Electronegativity is the tendency of an atom to attract shared pair of electrons towards itself in a covalent bond.

Down the group, electronegativity decreases because atomic size increases and shielding effect increases. Therefore, the nucleus attracts bonding electrons less strongly.

Example:



OR: Volume of 4.5 g H_2 at STP

Molar mass of $H_2 = 2g/mol$

$$moles = \frac{4.5}{2} = 2.25$$

At STP:

$$1mol = 22.4dm^3$$

$$Volume = 2.25 \times 22.4 = 50.4dm^3$$

Answer: $50.4dm^3$

Q2(xiii)

Main: Vapour pressure

Vapour pressure is the pressure exerted by vapours of a liquid in equilibrium with its liquid at a given temperature.

Given vapour pressure:



Reason: stronger intermolecular forces give lower vapour pressure.

Water has strong hydrogen bonding, so lowest vapour pressure.

Ethanol also has hydrogen bonding but weaker than water.

Pentane has only London dispersion forces, so it evaporates easily and has highest vapour pressure.

OR: pH of buffer

Given:

$$\begin{aligned}[HCOOH] &= 0.11M \\ [HCOONa] &= 0.11M \\ pK_a &= 3.76\end{aligned}$$

Henderson equation:

$$\begin{aligned}pH &= pK_a + \log \frac{[salt]}{[acid]} \\ pH &= 3.76 + \log \frac{0.11}{0.11} \\ pH &= 3.76 + \log 1 \\ pH &= 3.76\end{aligned}$$

Answer: pH = 3.76

Section C — Long Questions

Q3

Main: First-order reaction

Given:

$$[A]_0 = 0.10 \text{ mol dm}^{-3}$$
$$\text{Rate} = 2.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$$

For first order:

$$\text{Rate} = k[A]$$
$$k = \frac{\text{Rate}}{[A]}$$
$$k = \frac{2.0 \times 10^{-3}}{0.10}$$
$$k = 2.0 \times 10^{-2} \text{ s}^{-1}$$

Using half-life method:

$$t_{1/2} = \frac{0.693}{k}$$
$$k = \frac{0.693}{100}$$
$$k = 6.93 \times 10^{-3} \text{ s}^{-1}$$

Answer:

By initial rate method:

$$k = 2.0 \times 10^{-2} \text{ s}^{-1}$$

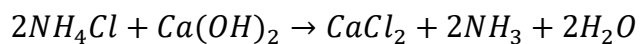
By half-life method:

$$k = 6.93 \times 10^{-3} \text{ s}^{-1}$$

These values are different, so the given data are **not fully consistent**.

OR: Ammonia from ammonium chloride

Reaction:



Given:

$$\begin{aligned}NH_4Cl &= 200g \\ Ca(OH)_2 &= 100g\end{aligned}$$

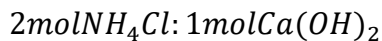
Molar masses:

$$\begin{aligned}NH_4Cl &= 53.5g/mol \\ Ca(OH)_2 &= 74g/mol\end{aligned}$$

Moles:

$$\begin{aligned}n(NH_4Cl) &= \frac{200}{53.5} = 3.738mol \\ n(Ca(OH)_2) &= \frac{100}{74} = 1.351mol\end{aligned}$$

According to equation:

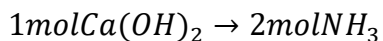


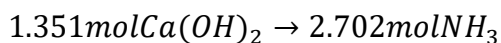
For 1.351mol $Ca(OH)_2$, required NH_4Cl :

$$2 \times 1.351 = 2.702mol$$

Available $NH_4Cl = 3.738mol$, so NH_4Cl is excess and $Ca(OH)_2$ is limiting.

From equation:





Mass of NH_3 :

$$2.702 \times 17 = 45.93 \text{ g}$$

Excess NH_4Cl left:

$$3.738 - 2.702 = 1.036 \text{ mol}$$

$$\text{mass} = 1.036 \times 53.5 = 55.43 \text{ g}$$

Answers:

Ammonia produced:

$$45.93 \text{ g}$$

Excess NH_4Cl left:

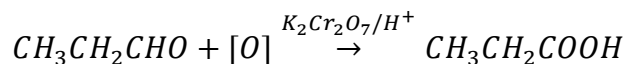
$$55.43 \text{ g}$$

Q4

Main: Preparation of propanoic acid

a) From propanal

Propanal is oxidized to propanoic acid.

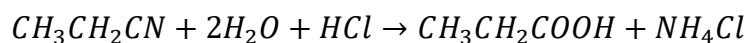


Reagent: acidified $\text{K}_2\text{Cr}_2\text{O}_7$ or KMnO_4

Condition: heat/reflux.

b) From propanenitrile

Nitriles give carboxylic acids by acidic hydrolysis.

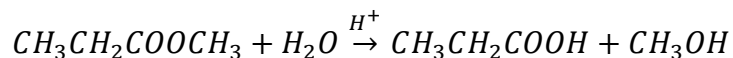


Reagent: dilute acid, HCl/H_2O

Condition: reflux.

c) From methyl propanoate

Ester hydrolysis gives carboxylic acid.

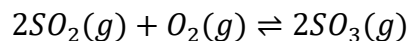


Reagent: dilute acid

Condition: reflux.

OR: Increasing yield of sulphur trioxide

Reaction:



This is exothermic:

$$\Delta H < 0$$

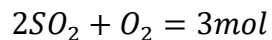
1) Low temperature

Forward reaction is exothermic, so low temperature favors SO_3 formation.

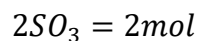
But very low temperature makes reaction slow, so about $450^\circ C$ is used.

2) High pressure

Reactant side has 3 moles of gas:



Product side has 2 moles of gas:



High pressure favors the side with fewer gas moles, so yield of SO_3 increases.

3) Remove SO_3

Removing SO_3 shifts equilibrium forward according to Le Chatelier's principle, increasing yield.

Important note

A catalyst V_2O_5 increases rate but does **not** increase equilibrium yield directly.

Q5

Main: Isomers of $C_4H_{10}O$

Alcohols with formula $C_4H_{10}O$ have four structural isomers.

1) Butan-1-ol



Primary alcohol.

2) Butan-2-ol



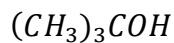
Secondary alcohol.

3) 2-methylpropan-1-ol



Primary alcohol.

4) 2-methylpropan-2-ol



Tertiary alcohol.

Type of isomerism

They show:

1. Chain isomerism

Straight chain and branched chain structures.

Example: butan-1-ol and 2-methylpropan-1-ol.

2. Position isomerism

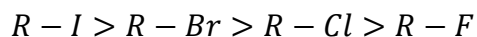
-OH group changes position.

Example: butan-1-ol and butan-2-ol.

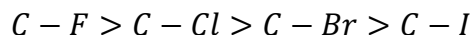
OR

a) Reactivity of alkyl halides

Order of reactivity:



Reason: reactivity depends mainly on carbon-halogen bond strength.

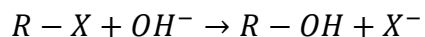


The C-I bond is weakest, so it breaks most easily. Therefore alkyl iodide is most reactive.

b) Test for halogens in alkyl halides

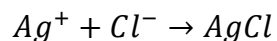
Alkyl halide is first warmed with alcoholic silver nitrate or hydrolysed with NaOH and then tested with $AgNO_3$.

General reaction:



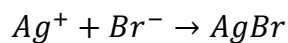
Then:

Chloride test



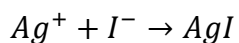
White precipitate.

Bromide test



Cream precipitate.

Iodide test

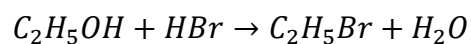


Yellow precipitate.

Q6

Main: Ethanol reactions

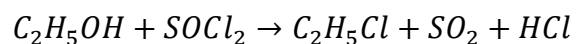
1) Ethanol with HBr



Product: bromoethane

Condition: heat with concentrated HBr.

2) Ethanol with $SOCl_2$

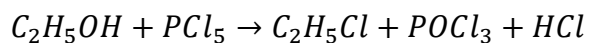


Product: chloroethane

Condition: room temperature or mild heating, often in pyridine.

This reaction is useful because gaseous by-products escape, making reaction easier.

3) Ethanol with PCl_5



Product: chloroethane

Condition: dry conditions.

OR: Hybridization, bond angle and shape

a) PH_3

Phosphorus has 5 valence electrons.

It forms 3 sigma bonds with hydrogen and has 1 lone pair.

Steric number:

$$3 + 1 = 4$$

Hybridization:



Electron pair geometry: tetrahedral

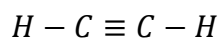
Molecular shape: trigonal pyramidal

Bond angle: about 93.5°

Reason: lone pair-bond pair repulsion compresses bond angle.

b) C_2H_2

Structure:



Each carbon forms:

1 sigma bond with H

1 sigma bond with other C

2 pi bonds between carbons

Steric number around each carbon = 2

Hybridization:



Shape:

linear

Bond angle:

180°